1 Solution — GIAM 7.2 Problem 2

1.1 Problem

Complete the proof of the fact that "Every graph has an even number of odd nodes."

1.2 Setup and Notation

Let $G = (V, E)$ be a finite (multi-)graph: V is its vertex set ("nodes") and E its multiset of edges (parallel edges and self-loops allowed). For each $v \in V$, the *degree* $\deg(v)$ is the number of edge–incidences at v, where a self-loop $\{v, v\}$ counts as 2 incidences at v (entering and leaving).

A vertex is called *odd* if $\deg(v)$ is odd, and *even* otherwise. Write

$$V_{\text{odd}} \;=\; \{\, v \in V \;:\; \deg(v)\ \text{is odd}\,\}, \qquad V_{\text{even}} \;=\; V \setminus V_{\text{odd}}.$$

Claim. $|V_{\text{odd}}|$ is even.

The proof has three small steps: a *degree-sum identity*, a *parity decomposition* of that sum, and the *parity of a sum of odd numbers*.

1.3 Step 1 — The Degree-Sum Identity

$$\sum_{v \in V} \deg(v) \;=\; 2\,|E|.$$

Why. Count *incidences* — that is, ordered pairs (e, v) such that v is an endpoint of e — in two ways:

- **By vertex.** Each $v \in V$ is incident to exactly $\deg(v)$ edges (counted with multiplicity for self-loops), so summing over vertices gives $\sum_v \deg(v)$.
- **By edge.** Each edge has exactly 2 incidences: a non-loop edge $\{u, w\}$ with $u \neq w$ contributes one incidence at u and one at w; a self-loop $\{v, v\}$ contributes two incidences at v. Summing over edges gives $2\,|E|$.

The two counts equal the same quantity, which proves the identity. $\square$

This identity already appeared informally in Chapter 7.1 Problem 9 (the handshake count $\binom{n}{2}$), where it was used in the form "$n(n-1) = 2 \cdot \#\text{handshakes}$".

1.4 Step 2 — Decompose the Sum by Parity

Partition the sum from Step 1 along the partition $V = V_{\text{even}} \sqcup V_{\text{odd}}$:

$$\underbrace{\sum_{v \in V} \deg(v)}_{=2|E|\ \text{(Step 1)}} \;=\; \sum_{v \in V_{\text{even}}} \deg(v) \;+\; \sum_{v \in V_{\text{odd}}} \deg(v).$$

Two observations:

- The left-hand side $2|E|$ is **even** — it is literally twice an integer.
- The first sum on the right is a sum of *even* numbers, hence also **even**.

Subtracting the second observation from the first, the remaining sum on the right satisfies

$$\sum_{v \in V_{\text{odd}}} \deg(v) \;=\; \underbrace{2|E|}_{\text{even}} \;-\; \underbrace{\sum_{v \in V_{\text{even}}} \deg(v)}_{\text{even}} \quad\Longrightarrow\quad \sum_{v \in V_{\text{odd}}} \deg(v)\ \text{ is even.}$$

1.5 Step 3 — Parity of a Sum of Odd Numbers

Lemma. A sum of k odd integers is even iff k is even.

Proof. Every odd integer is $\equiv 1 \pmod 2$. Reducing mod 2, a sum of k such integers is $\equiv \underbrace{1+1+\cdots+1}_{k} = k \pmod 2$. So the sum is even iff $k \equiv 0 \pmod 2$. $\square$

Applying this lemma to $\sum_{v \in V_{\text{odd}}} \deg(v)$ — a sum of $|V_{\text{odd}}|$ odd numbers — and combining with Step 2:

$$\underbrace{\sum_{v \in V_{\mathrm{odd}}} \deg(v)}_{\text{even, by Step 2}} \equiv |V_{\mathrm{odd}}| \pmod{2} \implies |V_{\mathrm{odd}}| \text{ is even.}$$

This is exactly the claim. ■

1.6 Sanity Checks on Small Graphs

Graph	**Degree sequence**	V_{odd}	**$ V_{\mathrm{odd}} $**
Empty graph on n vertices	$0, 0, \ldots, 0$	$\emptyset$	0
Path P_3 (three nodes in a row)	$1, 2, 1$	endpoints	2
Triangle K_3	$2, 2, 2$	$\emptyset$	0
Complete graph K_4	$3, 3, 3, 3$	all four	4
Complete graph K_5	$4, 4, 4, 4, 4$	$\emptyset$	0
Bipartite $K_{3,3}$	$3, 3, 3, 3, 3, 3$	all six	6
Königsberg bridge graph	$3, 3, 3, 5$	all four	4

Table 1.1

Each entry has $|V_{\mathrm{odd}}|$ even — including Königsberg, which has *four* odd vertices, ruling out both an Eulerian circuit (needs 0) and an Eulerian path (needs exactly 2). The lemma is consistent with — and indeed *forces* — the Königsberg impossibility.

1.7 Remark — The "Handshaking" Interpretation

The result is often stated colloquially as the **handshaking lemma**:

> *In any gathering, the number of people who have shaken hands an odd number of times is even.*

The dictionary: people are vertices, handshake events are edges, and the number of handshakes a person has done is their degree. The identity $\sum \deg(v) = 2|E|$ is the formal statement that "each handshake is felt by exactly two hands."

This is the *structural* counterpart of the *quantitative* handshake count from Chapter 7.1 Problem 9 (which computed $\binom{n}{2}$ total handshakes in a full meet-and-greet). Both follow from the same degree-sum identity — used here for its parity, there for its magnitude.

1.8 Remark — Why "Complete the Proof"

The textbook poses this exercise as *completing* the proof, presumably because the heart of the argument is the degree-sum identity (Step 1), which the section establishes as motivation. Steps 2 and 3 — partitioning by parity and noting that a sum of k odd numbers has parity k — are what remains to be written down explicitly. The argument's elegance is in how a quantitative identity ($\sum \deg(v) = 2|E|$) is parlayed into a qualitative parity result ($|V_{\text{odd}}|$ is even) via a simple counting-mod-2 observation.